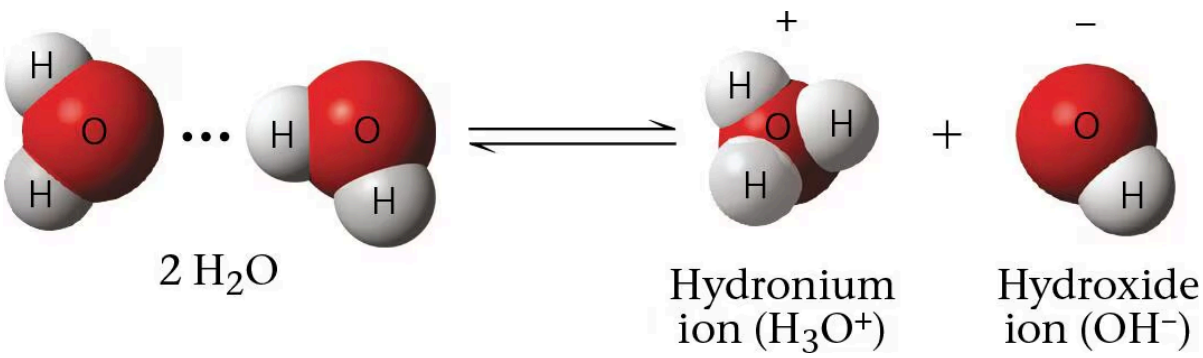
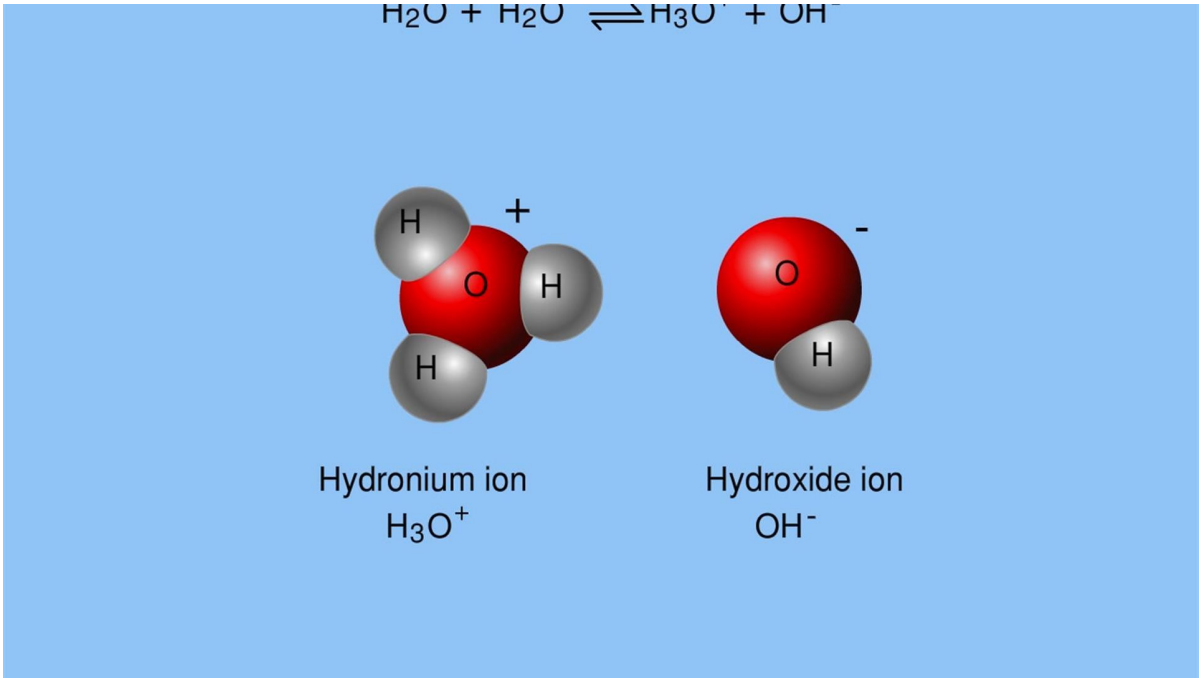
3.3: Acidic and basic conditions affect living organisms

Occasionally, a hydrogen atom participating in a hydrogen bond between two water molecules shifts from one molecule to the other. When this happens, the hydrogen atom leaves its electron behind, and what is actually transferred is a **hydrogen ion** (H+), a single proton with a charge of 1+. The water molecule that lost a proton is now a **hydroxide ion** (OH−), which has a charge of 1−. The proton binds to the other water molecule, making that molecule a **hydronium ion** (H3O+). We can picture the chemical reaction as follows:



Animation

Dissociation of Water Molecules



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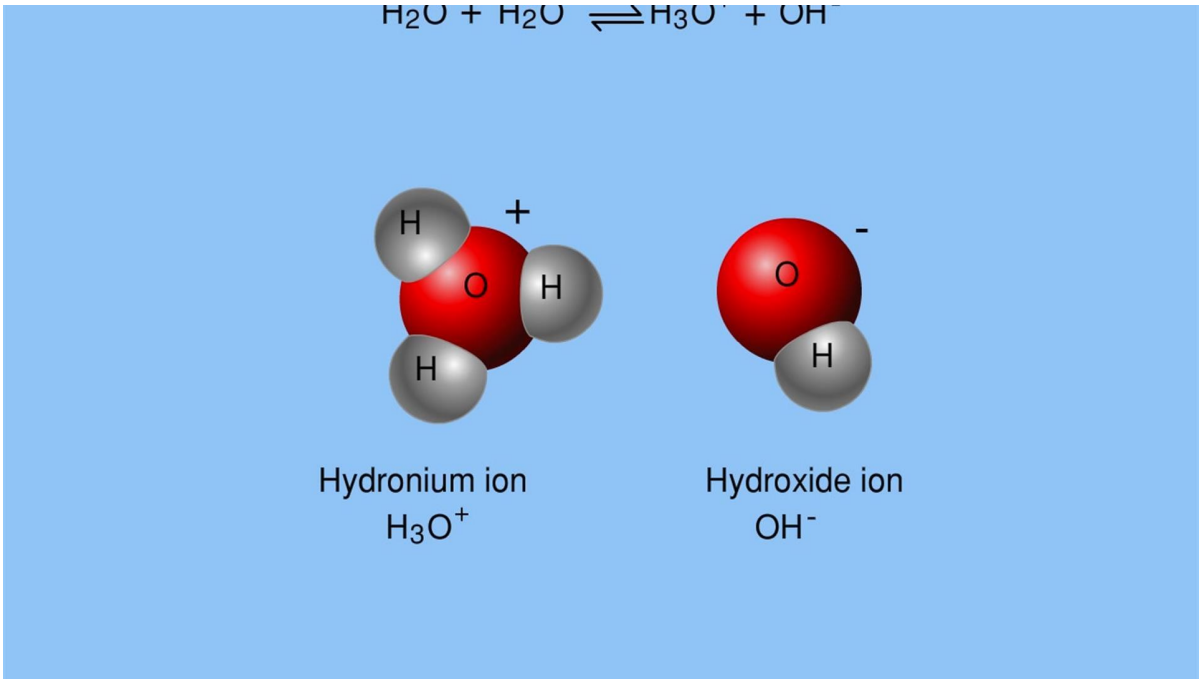

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By convention, H+ (the hydrogen ion) is used to represent H3O+ (the hydronium ion), and we follow that practice in this book. Keep in mind, though, that H+ does not exist on its own in an aqueous solution. It is always associated with a water molecule in the form of H3O+.

As indicated by the double arrows, this is a reversible reaction that reaches a state of dynamic equilibrium when water molecules dissociate at the same rate that they are being reformed from H+ and OH−. At this equilibrium point, the concentration of water molecules greatly exceeds the concentrations of H+ and OH−. In pure water, only one water molecule in every 554 million is dissociated; the concentration of H+ and of OH− in pure water is therefore 10−7M (at 25°C). This means there is only one ten-millionth of a mole of hydrogen ions per liter of pure water and an equal number of hydroxide ions. (Even so, this is a huge number —over 60,000trillion—of each ion in a liter of pure water.)

Although the dissociation of water is reversible and statistically rare, it is exceedingly important in the chemistry of life. H+ and OH− are very reactive. Changes in their concentrations can drastically affect a cell's proteins and other complex molecules. As we have seen, the concentrations of H+ and OH− are equal in pure water, but adding certain kinds of solutes, called acids and bases, disrupts this balance. Biologists use

something called the pH scale to describe how acidic or basic (the opposite of acidic) a solution is. In the remainder of this chapter, you will learn about acids, bases, and pH and why changes in pH can adversely affect organisms.

Acids and Bases

What would cause an aqueous solution to have an imbalance in H+ and OH− concentrations? When acids dissolve in water, they donate additional H+ to the solution. An **acid** is a substance that increases the hydrogen ion concentration of a solution. For example, when hydrochloric acid (HCl) is added to water, hydrogen ions dissociate from chloride ions:

$HCl \rightarrow H^+ + Cl^-$

This source of H+ (dissociation of water is the other source) results in an acidic solution—one having more H+ than OH−.

A substance that reduces the hydrogen ion concentration of a solution is called a **base**. Some bases reduce the H+ concentration directly by accepting hydrogen ions. Ammonia (NH_3), for instance, acts as a base when the unshared electron pair in nitrogen's valence shell attracts a hydrogen ion from the solution, resulting in an ammonium ion (NH_4^+):

$NH_3 + H^+ \rightleftharpoons NH_4^+$

Other bases reduce the H+ concentration indirectly by dissociating to form hydroxide ions, which combine with hydrogen ions and form water. One such base is sodium hydroxide (NaOH), which in water dissociates into its ions:

$NaOH \rightarrow Na^+ + OH^-$

In either case, the base reduces the H+ concentration. Solutions with a higher concentration of OH− than H+ are known as basic solutions. A solution in which the H+ and OH− concentrations are equal is said to be neutral.

Notice that single arrows were used in the reactions for HCl and NaOH. These compounds dissociate completely when mixed with water, so hydrochloric acid is called a strong acid and sodium hydroxide a strong base. In contrast, ammonia is a weak base. The double arrows in the reaction for ammonia indicate that the binding and release of hydrogen ions are reversible reactions, although at equilibrium there will be a fixed ratio of NH_4^+ to NH_3.

Weak acids are acids that reversibly release and accept back hydrogen ions. An example is carbonic acid:

H_2CO_3 Carbonic acid $\rightleftharpoons HCO_3^-$ Bicarbonate ion $+ H^+$ Hydrogen ion

Here the equilibrium so favors the reaction in the left direction that when carbonic acid is added to pure water, only 1% of the molecules are dissociated at any particular time. Still, that is enough to shift the balance of H+ and OH− from neutrality.

The pH Scale

In any aqueous solution at 25°C, the product of the H+ and OH− concentrations is constant at 10−14. This can be written

$$[H^+][OH^-]=10^{-14}$$

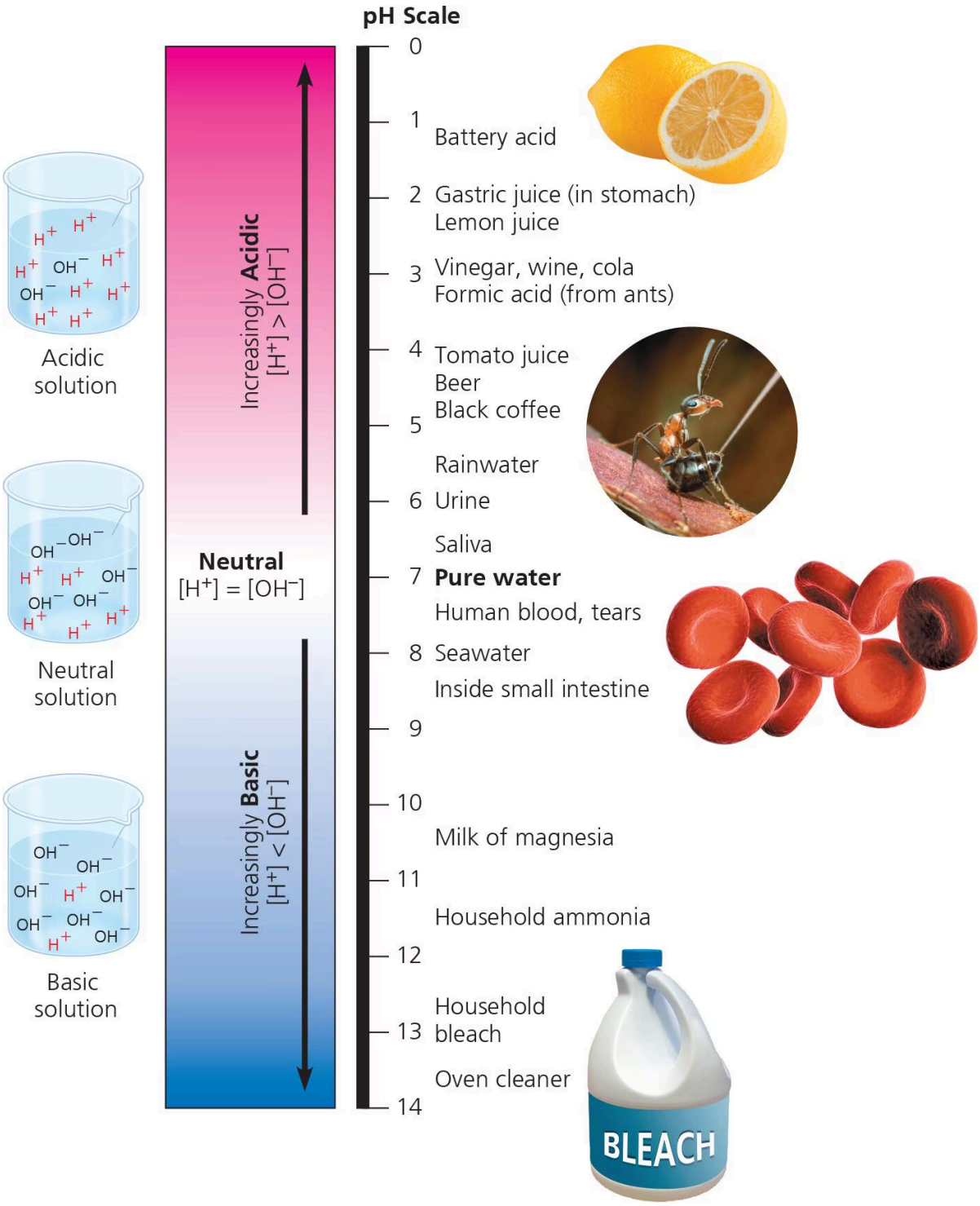
(The brackets indicate molar concentration.) As previously mentioned, in a neutral solution at 25°C, $[H^+]=10^{-7}$ and $[OH^-]=10^{-7}$. Therefore, the product of [H+] and [OH−] in a neutral solution at 25°C is 10−14. If enough acid is added to a solution to increase [H+] to 10−5M, then [OH−] will decline by an equivalent factor to 10−9M (note that $10^{-5}\times10^{-9}=10^{-14}$). This constant relationship expresses the behavior of acids and bases in an aqueous solution. An acid not only adds hydrogen ions to a solution, but also removes hydroxide ions because of the tendency for H+ to combine with OH−, forming water. A base has the opposite effect, increasing OH− concentration but also reducing H+ concentration by the formation of water. If enough of a base is added to raise the OH− concentration to 10−4M, it will cause the H+ concentration to drop to 10−10M. Whenever we know the concentration of either H+ or OH− in an aqueous solution, we can deduce the concentration of the other ion.

The pH scale **(Figure 3.11)** is a simple numerical method for expressing the range of H+ concentrations. The H+ concentrations of solutions can vary by a factor of 100 trillion or more. Instead of using moles per liter, the pH scale compresses the range of H+ concentrations by employing logarithms. The **pH** of a solution is defined as the negative logarithm (base 10) of the H+ concentration:

$$pH=-\log [H^+]$$

Figure 3.11

The pH scale and pH values of some aqueous solutions.



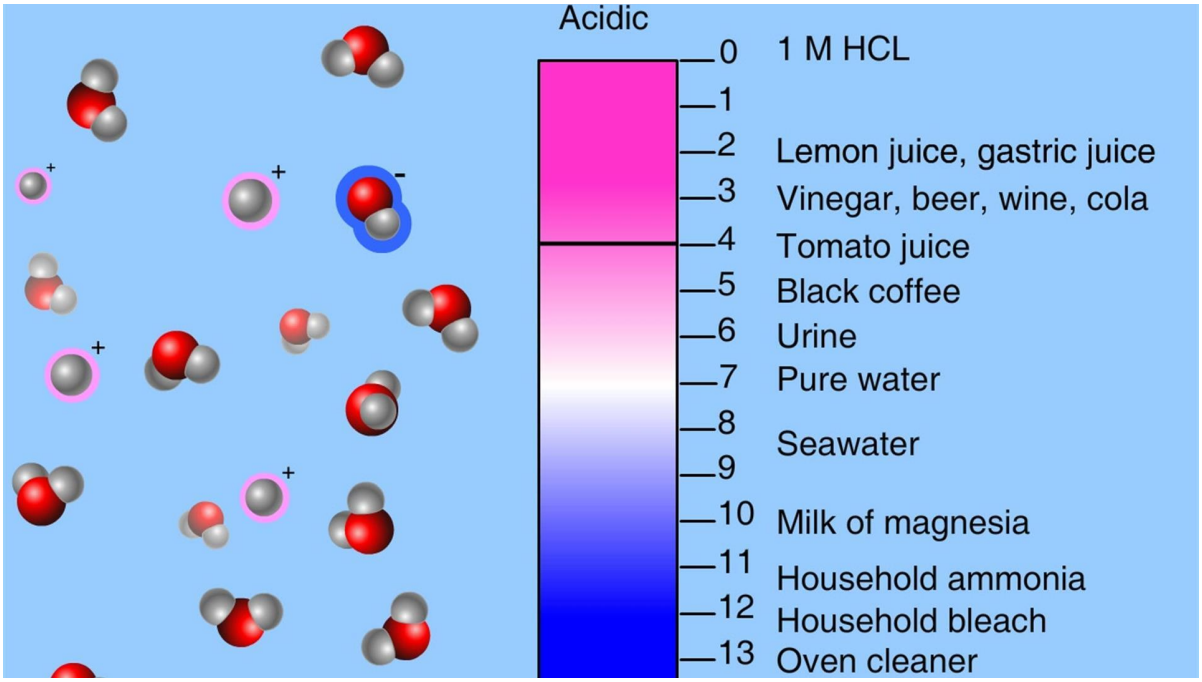
Increasingly acidic (number of H plus greater than O H minus) ranges from 6 to 0 p H. Acidic examples include Battery acid (p H = 1), Gastric juice, lemon juice (p H = 2), Vinegar, wine, cola (p H = 3), Tomato juice, Beer, Black Coffee (p H = 4), Rainwater (p H = 5.5), and urine (p H = 6). Neutral (number of H plus = O H minus) 7 p H.

Neutral and almost neutral examples include Saliva (p H = 6.5), Pure water (p H = 7), Human blood, tears (p H = 7.5). Increasingly Basic (number of H plus less than O H minus) 8 – 14 p H.

Basic examples include Seawater (p H = 8), inside small intestine (p H = 8.5), Milk of magnesia (p H = 10.5), Household ammonia (p H = 11.5), Household bleach (p H = 12.5), and Oven cleaner (p H = 13.5).

Animation

Acids, Bases, and pH



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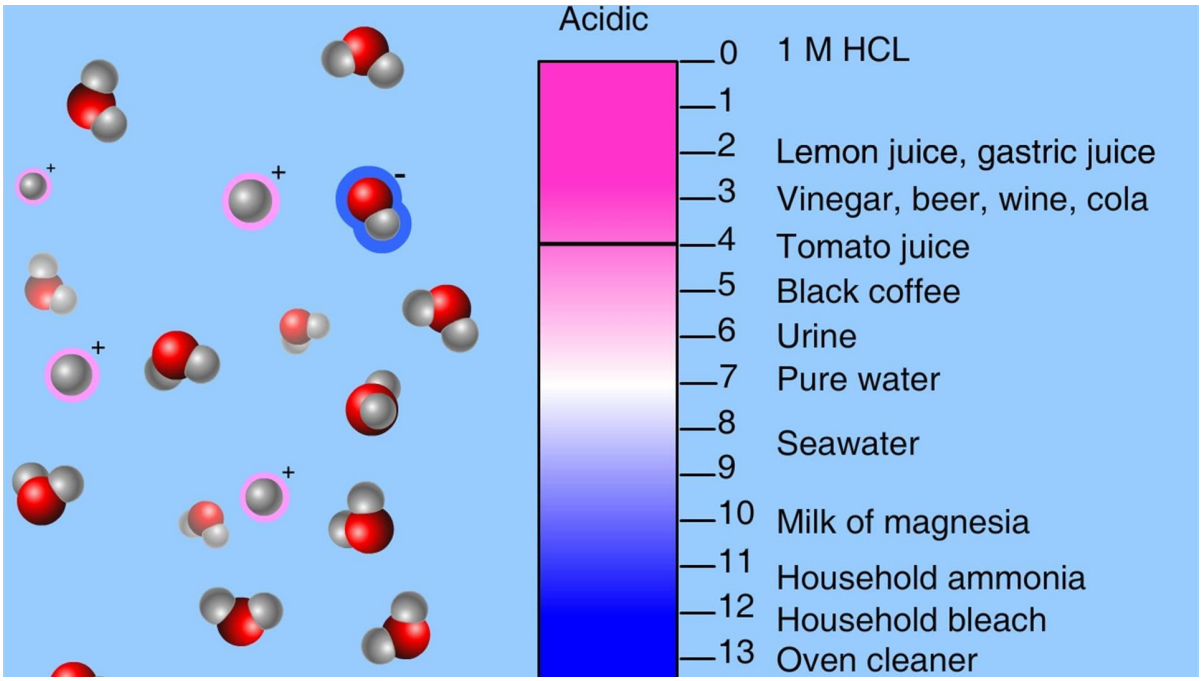

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For a neutral aqueous solution, [H+] is 10−7M, giving us

−log 10−7=−(−7)=7

Notice that pH *decreases* as H+ concentration *increases* (see Figure 3.11). Notice, too, that although the pH scale is based on H+ concentration, it also implies OH− concentration. A solution of pH 10 has a hydrogen ion concentration of 10−10M and a hydroxide ion concentration of 10−4M.

The pH of a neutral aqueous solution at 25°C is 7, the midpoint of the pH scale. A pH value less than 7 denotes an acidic solution; the lower the number, the more acidic the solution. The pH for basic solutions is above 7. Most biological fluids, such as blood and saliva, are within the range of pH 6–8. There are a few exceptions, however, including the strongly acidic digestive juice of the human stomach (gastric juice), which has a pH of about 2.

Remember that each pH unit represents a tenfold difference in H+ and OH− concentrations. It is this mathematical feature that makes the pH scale so compact. A solution of pH 3 is not twice as acidic as a solution of pH 6, but 1,000times(10×10×10) more acidic. When the pH of a solution changes slightly, the actual concentrations of H+ and OH− in the solution change substantially.

Buffers

The internal pH of most living cells is close to 7. Even a slight change in pH can be harmful because the chemical processes of the cell are very sensitive to the concentrations of hydrogen and hydroxide ions. The pH of human blood is very close to 7.4, which is slightly basic. A person cannot survive for more than a few minutes if the blood pH drops to 7 or rises to 7.8, and a chemical system exists in the blood that maintains a stable pH. If 0.01 mol of a strong acid is added to a liter of pure water, the pH drops from 7.0 to 2.0. If the same amount of acid is added to a liter of blood, however, the pH decrease is only from 7.4 to 7.3. Why does the addition of acid have so much less of an effect on the pH of blood than it does on the pH of water?

The presence of substances called buffers allows biological fluids to maintain a relatively constant pH despite the addition of acids or bases. A **buffer** is a substance that minimizes changes in the concentrations of H+ and OH− in a solution. It does so by accepting hydrogen ions from the solution when they are in excess and donating hydrogen ions to the solution when they have been depleted. Most buffer solutions contain a weak acid and its corresponding base, which combine reversibly with hydrogen ions.

Several buffers contribute to pH stability in human blood and many other biological solutions. One of these is carbonic acid (H_2CO_3), which is formed when CO_2 reacts with water in blood plasma. As mentioned earlier, carbonic acid dissociates to yield a bicarbonate ion (HCO_3^-) and a hydrogen ion (H+):

H_2CO_3H+ donor (acid)⇌Response toa drop in pHResponse to a rise in pHHCO_3^-
H+ acceptor(base)+H+Hydrogenion

The chemical equilibrium between carbonic acid and bicarbonate acts as a pH regulator, the reaction shifting left or right as other processes in the solution add or remove hydrogen ions. If the H+ concentration in blood begins to fall (that is, if pH rises), the reaction proceeds to the right and more carbonic acid dissociates, replenishing hydrogen ions. But when the H+ concentration in blood begins to rise (when pH drops), the reaction proceeds to the left, with HCO_3^- (the base) removing the hydrogen ions from the solution and forming H_2CO_3. Thus, the carbonic acid–bicarbonate buffering system consists of an acid and a base in equilibrium with each other. Most other buffers are also acid-base pairs.

Acidification: A Threat to Our Oceans

Among the many threats to water quality posed by human activities is the burning of fossil fuels, which releases CO_2 into the atmosphere. The resulting increase in atmospheric CO_2 levels has caused global warming and other aspects of climate change (see Concept 56.4). In addition, about 25% of human-generated CO_2 is absorbed by the oceans. In spite of the huge volume of water in the oceans, scientists worry that the absorption of so much CO_2 will harm marine ecosystems.

Recent data have shown that such fears are well founded. When CO_2 dissolves in seawater, it reacts with water to form carbonic acid, which lowers ocean pH. This process, known as **ocean acidification**, alters the delicate balance of conditions for life in the oceans **(Figure 3.12)**. Based on measurements of the CO_2 level in air bubbles trapped in ice over thousands of years, scientists calculate that the pH of the oceans is 0.1 pH unit lower (more acidic) now than at any time in the past 420,000years. Recent studies predict that it will drop another 0.3–0.5 pH unit by the end of this century.

Figure 3.12

Atmospheric CO_2 from human activities and its fate in the ocean.

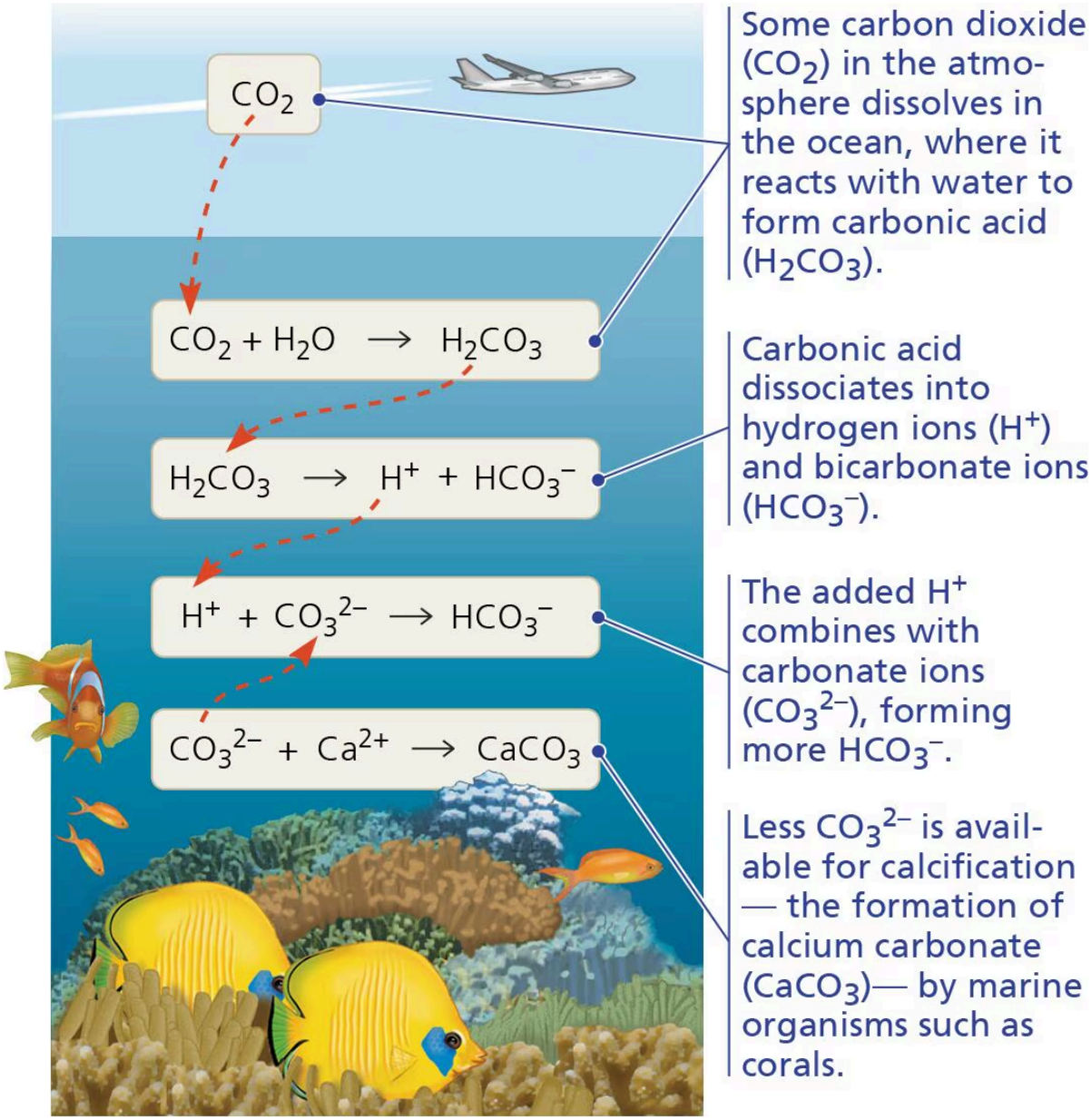


Illustration shows an ocean with an airplane flying in the sky above. There is a streak of exhaust coming out of the rear of the airplane. A label showing carbon dioxide (C O subscript 2) in the sky. A dotted arrow points down to the first step.

- Some carbon dioxide (C O subscript 2) in the atmosphere dissolves in the ocean where it reacts with water (H subscript 2 O) to form carbonic acid (H subscript 2 C O subscript 3). Equation: carbon dioxide (C O subscript 2) plus water (H subscript 2 O) yields carbonic acid (H subscript 2 C O subscript 3). A dotted arrow points down to the next step.
- Carbonic acid dissociates into hydrogen ions (H superscript plus) and bicarbonate ions (H C O subscript 3 minus). Equation: carbonic acid (H subscript 2 C O subscript 3) yields hydrogen ions

(H superscript plus) plus bicarbonate ions (H C O subscript 3 superscript minus). A dotted arrow points down to the next step.

- The added hydrogen (H plus) combines with carbonate ions (H C O subscript 3 superscript 2 minus) forming more bicarbonate ions (H C O subscript 3 minus). Equation: hydrogen (H plus) plus carbonate (H C O subscript 3 superscript 2 minus) yields bicarbonate ions (H C O subscript 3 minus). A dotted arrow points up from this step to the preceding step.
- Less carbonate (H C O subscript 3 superscript 2 minus) is available for calcification—the formation of calcium carbonate (C a C O subscript 3)—by marine organisms such as corals. Equation: carbonate (H C O subscript 3 superscript 2 minus) plus calcium (C a superscript 2 plus) yields calcium carbonate (C a C O subscript 3).

VISUAL SKILLS *Summarize the effect of adding excess* CO_2 *to the oceans on the calcification process in the final equation.*

*For suggested answer, see**Appendix A.*

As seawater acidifies, the extra hydrogen ions combine with carbonate ions (CO_3^{2-}) to form bicarbonate ions (HCO_3^-), thereby reducing the carbonate ion concentration (see Figure 3.12). Scientists predict that ocean acidification will cause the carbonate ion concentration to decrease by 40% by the year 2100. This is of great concern because carbonate ions are required for calcification, the production of calcium carbonate ($CaCO_3$) by many marine organisms, including reef-building corals and animals that build shells. The **Scientific Skills Exercise** allows you to work with data from an experiment examining the effect of carbonate ion concentration on coral reefs, using an artificial system. In 2018, researchers carried out the first CO_2 enhancement study on an unconfined natural coral reef, observing that addition of CO_2 suppressed calcification and concluding that ocean acidification is likely to cause "profound, ecosystem-wide changes in coral reefs." Coral reefs are sensitive ecosystems that act as havens for a great diversity of marine life. The disappearance of coral reef ecosystems would be a tragic loss of biological diversity.

Scientific Skills Exercise

Interpreting a Scatter Plot with a Regression Line

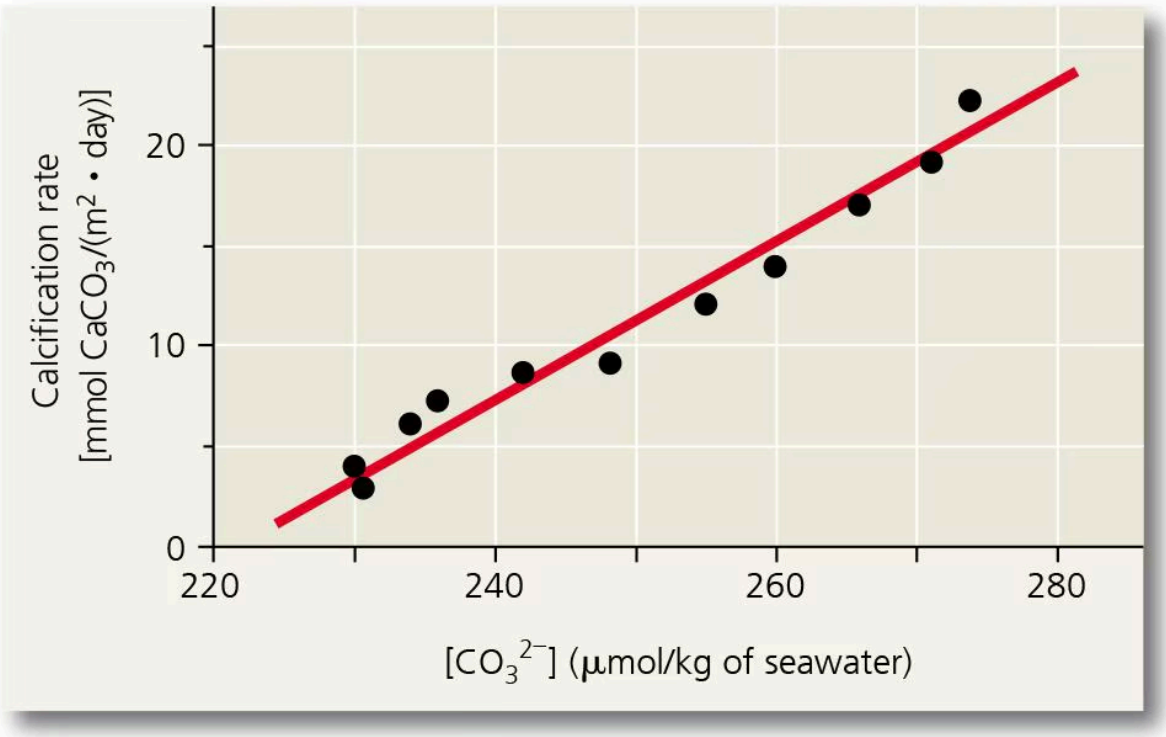
AP Science Practices 1, 2, 4, 5

How Does the Carbonate Ion Concentration of Seawater Affect the Calcification Rate of a Coral Reef? Scientists predict that acidification of the ocean due to higher levels of atmospheric CO_2 will lower the concentration of dissolved carbonate ions, which living corals use to build calcium carbonate reef

structures. In this exercise, you will analyze data from a controlled experiment that examined the effect of carbonate ion concentration ([CO_3^{2-}]) on calcium carbonate deposition, a process called calcification.

How the Experiment Was Done For several years, scientists conducted research on ocean acidification using a large coral reef aquarium at Biosphere 2 in Arizona. They measured the rate of calcification by the reef organisms and examined how the calcification rate changed with differing amounts of dissolved carbonate ions in the seawater.

Data from the Experiment The black data points in the graph form a scatter plot. The red line, known as a linear regression line, is the best-fitting straight line for these points.



Data from C. Langdon et al., Effect of calcium carbonate saturation state on the calcification rate of an experimental coral reef, *Global Biogeochemical Cycles* 14:639–654 (2000).

Linear regression line goes through (235, 5) and (260, 15). Calcification rate (millimole C a C O 3 per meter squared per day (y-axis)) versus Carbonate ion concentration (C O 3 superscript 2 minus micromole per kilogram of seawater (x-axis)). Linear regression line goes through (235, 5) and (260, 15).

INTERPRET THE DATA

- When presented with a graph of experimental data, the first step in analysis is to determine what each axis represents.
- In words, what is shown on the x-axis? (Include the units.)
- What is on the y-axis?
- Which variable is the independent variable—the one that was *manipulated* by the researchers?
- Which is the dependent variable—the one that responded to or depended on the treatment, which was *measured* by the researchers? (For additional information about graphs, see the Scientific Skills Review in Appendix D.)
- Based on the data shown in the graph, describe in words the relationship between carbonate ion concentration and calcification rate.

- - If the seawater carbonate ion concentration is 270 μmol/kg, estimate the rate of calcification and how many days it would take 1 square meter of reef to accumulate 30 mmol of calcium carbonate ($CaCO_3$).
- If the seawater carbonate ion concentration is 250 μmol/kg, what is the approximate rate of calcification, and approximately how many days would it take 1 square meter of reef to accumulate 30 mmol of calcium carbonate?
- If the carbonate ion concentration decreases, how does the calcification rate change, and how does that affect the time it takes coral to grow?
- - Which step of the process in Figure 3.12 is measured in this experiment?
- Are the results of this experiment consistent with the hypothesis that increased atmospheric [CO_2] will slow the growth of coral reefs? Why or why not?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

If there is any reason for optimism about the future quality of water resources on our planet, it is that we have made progress in learning about the delicate chemical balances in oceans, lakes, and rivers. Continued progress can come only from the actions of informed individuals, like yourselves, who are concerned about environmental quality. This requires understanding the crucial role that water plays in the suitability of the environment for continued life on Earth.